The Tail-Certificate Protocol

Heung-No Lee, *Senior Member, IEEE*
INFONET Laboratory, School of Electrical Engineering and Computer Science
Gwangju Institute of Science and Technology (GIST), Gwangju, Republic of Korea
Email: heungno@gist.ac.kr

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Abstract

We present the Tail-Certificate Protocol, an economic and mechanism-design layer that operates above the WorldLand consensus protocol [1] and resolves the long-standing tension between consensus participation and revenue-generating compute service in dual-mode blockchain networks. The central primitive is the *Tail Certificate*: a non-interactive, publicly verifiable proof that a given compute slot's coin toss for a specific block resulted in tails, and is therefore not subject to mining duty. By making valid Tail Certificates a precondition for receiving any AI workload through the on-chain marketplace, the protocol converts the choice between mining and service from a per-block economic decision—which a rational provider could exploit to defect from consensus—into a cryptographically enforced exclusion. We formalize a hybrid ticket model in which a provider's mining tickets equal $\min(\lfloor \text{stake}/S_0 \rfloor, \text{capacity})$, simultaneously bounding stake-only Sybil and hardware-only Sybil attacks. We introduce a hierarchical compute identity scheme (Provider DID, Machine DID, Compute Slot DID) operating at GPU granularity, a tiered identity model spanning attested data-center GPUs to consumer hardware, a three-part reward decomposition (winner, participation, availability), a dynamic security-pricing rule that rebalances incentives when consensus capacity drops below threshold, and a marketplace-revenue redirection that channels a portion of service fees into the consensus security pool. The combined mechanism eliminates the marketplace-layer mining-defection vulnerability in WorldLand and provides a self-stabilizing equilibrium across mining and service economies. The protocol is targeted for activation in the Rockies Upgrade v.1.2 milestone.

Index Terms

Blockchain, mechanism design, decentralized GPU marketplace, DePIN, Verifiable Random Function, dual-mode consensus, incentive compatibility, Sybil resistance, confidential computing.

I. Introduction

The WorldLand network occupies an unusual position in the design space of blockchain protocols: each node is required to participate in Proof-of-Work consensus only when probabilistically selected by a Verifiable Coin Toss (VCT), as formalized in the companion paper *The Coin-Toss Protocol* [1]. Outside the brief windows in which a node is selected, its compute capacity—typically a high-performance GPU—is free to serve revenue-generating workloads such as AI inference, model training, and rendering. This dual-mode operation underpins the network's claim to two-orders-of-magnitude energy reduction relative to Bitcoin while preserving Nakamoto-style asynchronous mining.

The dual-mode design, however, exposes a subtle but consequential vulnerability at the economic layer. When a node is selected for mining via VCT, performing the ECCPoW work consumes the same GPU that could otherwise be earning service revenue. If the spot price of compute exceeds the expected mining reward, a rational provider will be tempted to ignore the mining duty and continue serving the lucrative workload—defecting from consensus despite having received a valid mining ticket. Because VCT is, by construction, a private operation evaluated within the node, the rest of the network cannot directly observe which nodes were selected, complicating any direct slashing-based enforcement. Naive responses—making the mining duty publicly observable, requiring trusted attestation of the agent's behavior, or simply hoping providers behave honestly—all impose substantial costs in privacy, infrastructure complexity, or security margin.

This paper introduces a different approach. Rather than enforce mining duty by direct observation, we make the duty cryptographically incompatible with concurrent service revenue at the marketplace layer. The mechanism is the *Tail Certificate*: a publicly verifiable proof that a node's VCT outcome at a given block was tails, and that the node is therefore exempt from mining duty for that block. The on-chain AI workload broker requires a fresh Tail Certificate for each compute slot for each block in which that slot wishes to receive paid work. A node selected for mining (heads) cannot produce a valid Tail Certificate, so the marketplace will not assign it new workloads and will not release escrow payment for ongoing leases. The provider's only path to in-protocol revenue when selected is to perform the mining duty.

A. Contributions

This paper makes the following contributions:

1) We formalize the *Tail Certificate* primitive, define its construction and verification, and prove its unforgeability under standard VRF assumptions. We show how it integrates with an on-chain workload broker to convert mining defection into a non-event at the marketplace boundary.
2) We introduce a hierarchical compute identity scheme (Provider, Machine, Compute Slot) that operates at GPU granularity, allowing an eight-GPU server to mine on one slot while serving workloads on the other seven.
3) We define a hybrid ticket model, $\text{Tickets}_i = \min(\lfloor S_i/S_0 \rfloor,\ C_i)$, that simultaneously rules out stake-only Sybil and hardware-only Sybil attacks, and we analyze its incentive compatibility.
4) We propose a tiered compute-identity model that admits four classes of participating hardware—data-center GPUs with attestation, server-class hardware with TPM, consumer GPUs without strong attestation, and storage-only nodes—each with calibrated rights and stake requirements.
5) We design a three-part reward decomposition (winner reward, participation share reward, availability reward) and a service-revenue redirection scheme that channels a fraction of marketplace fees into a security pool funding availability and participation rewards.
6) We propose a dynamic security-pricing rule that rebalances incentives whenever active consensus capacity drops below a protocol-defined floor, ensuring self-stabilization of the dual-mode equilibrium.
7) We outline a phased deployment plan that aligns with the Rockies Upgrade roadmap of [1], with full marketplace activation targeted for Rockies Upgrade v.1.2.

B. Relation to the Coin-Toss Protocol

This paper is the second of a two-paper series on WorldLand's Rockies-era architecture. The first paper, *The Coin-Toss Protocol* [1], establishes the consensus protocol itself: the VCT function, the VRF-gated ECCPoW mining loop, the security analysis under hardware-anchored device identifiers

(DIDs), the formal treatment of 51% attack resistance, and the basic TPM-based DID construction. The present paper builds on that foundation. Cryptographic primitives, security assumptions, and the consensus state machine are taken from [1] as given, and we focus exclusively on the layers above consensus: identity granularity, ticket allocation, marketplace mechanics, reward design, and incentive equilibrium. Where the two papers describe overlapping behavior, we cite [1] rather than restate. The two protocols compose without modification.

C. Paper Organization

Section II reviews the necessary background. Section III defines the compute slot hierarchy and tiered identity model. Section IV introduces the hybrid ticket model. Section V formalizes Tail Certificates. Section VI describes the marketplace design. Section VII treats the three-part reward and dynamic pricing scheme. Section VIII analyzes provider strategy. Section IX outlines the implementation timeline. Sections X and XI discuss related work and conclude.

II. Background

A. The Coin-Toss Protocol in Brief

For each block at height h, every staked node i with VRF key sk_i evaluates

$$(y_i, \pi_i) = \mathrm{VRF}(sk_i,\ H_{h-1} \,\|\, h), \tag{1}$$

where H_{h-1} is the previous block header. If $y_i/2^{|y|} < p$ for the network-tunable parameter p, the node has obtained a heads result and is permitted—and required, under the present paper's economic enforcement—to attempt ECCPoW mining for block h. Otherwise, the node is on tails and must skip the block. Per-epoch dynamic adjustment maintains the expected number of heads per block at a target T, typically $T \in [10^2, 10^3]$. Security under this protocol, as proven in [1], requires the adversary to control more physical hardware-bound DIDs than the honest network in order to sustainedly produce a majority of blocks.

B. Decentralized GPU Marketplaces

A growing class of decentralized GPU marketplaces—Akash [9], Render [10], Clore [11], Golem [12], io.net [13], and others—allows providers to offer idle GPU capacity to remote tenants for a fee, settled through their respective tokens. These platforms generally operate as orderbooks or bidding systems, with workload assignment based on price, availability, and provider reputation. None, however, integrates such a marketplace with a Proof-of-Work consensus layer that draws on the same physical hardware. The unique problem this paper addresses—preventing providers from defecting from consensus duty toward higher-value service workloads—does not arise in their designs because their marketplaces do not coexist with mandatory mining duty.

C. Confidential Computing and Workload Verification

Recent generations of data-center GPUs (NVIDIA H100, H200, B100; AMD MI300) support hardware-rooted attestation and confidential computing modes [5]. These capabilities enable cryptographic proofs that a workload was executed on genuine, unmodified hardware. We do not require confidential computing for the basic Tail-Certificate construction, but optional integration in the upper participation tiers (Section III) enables verifiable AI workload markets that command premium pricing.

III. Compute Slot Hierarchy and Tiered Identity

A. Three-Level Identity Hierarchy

Real-world participating infrastructure ranges from single consumer GPUs to multi-GPU enterprise servers. Treating each server as a single identity, as is conventional in many blockchain protocols, would either underutilize multi-GPU systems (one mining duty preempts seven idle GPUs) or over-simplify Sybil accounting. We therefore decompose participating identity into three levels:

Definition 1 (Identity Hierarchy)**.** *A WorldLand participant maintains three classes of decentralized identifier:*

- ***Provider DID****: the legal or pseudonymous operator entity that bonds stake and enters into protocol-level agreements.*
- ***Machine DID****: a physical host (server, workstation, or single-GPU appliance) bound to a Provider DID, attested via TPM 2.0 endorsement keys following [1].*
- ***Compute Slot DID****: a single normalized compute unit—typically one GPU or one accelerator—bound to a Machine DID. The protocol's smallest unit of mining and serving granularity.*

VCT evaluation, mining duty, and Tail Certificate issuance all occur at the Compute Slot level. This permits an eight-GPU server in which one slot is selected to mine while the other seven simultaneously serve AI workloads—a critical efficiency property for capital-intensive infrastructure.

B. Tiered Compute Identity

Not all hardware is equally suited to consensus participation. Consumer GPUs lack the on-die attestation roots present in data-center accelerators, and require correspondingly stronger economic and behavioral anti-Sybil measures. We define four tiers (Table I).

TABLE I
Tiered Compute Identity

Tier	Hardware class	Attestation requirement	Consensus rights
A	H100, H200, B100/B200 (and equivalents)	GPU on-die RoT + TPM 2.0 + verifier signature	Full mining and service
B	Server-class GPU + TPM 2.0 host	TPM EK chain + benchmark fingerprint + liveness audit	Mining (capped) and service
C	Consumer GPU (RTX 4060, RTX 5090, etc.)	Stake + benchmark + reputation only	Service only; no mining initially
S	Storage-class node (no GPU)	Stake + retrieval audit	Storage service only; no mining

Tier C nodes participate in the marketplace as service providers but do not contribute to consensus security; they are excluded from the eligible mining set. Tier S nodes participate exclusively in storage workloads, independent of the VCT lottery. Tiers A and B contribute to consensus security with progressively stronger attestation requirements. Promotion between tiers occurs through governance-approved attestation upgrades; demotion is automatic upon attestation failure.

IV. The Hybrid Ticket Model

A. Motivation

A pure stake-weighted ticket allocation rewards token wealth without regard to physical compute, while a pure hardware-weighted allocation neglects the economic bonding that makes attacks costly. Each pure model fails differently: stake-only allocations let a wealthy adversary acquire mining influence without contributing actual work, while hardware-only allocations let well-resourced node operators dominate without economic skin in the game.

B. Definition

Definition 2 (Hybrid Ticket Allocation)**.** *Let provider i control bonded stake S_i (in WL tokens) and aggregate normalized capacity C_i (in compute units, defined as the sum of capacity weights c_j over the provider's registered Compute Slot DIDs j). Given protocol parameter S_0 (stake required per ticket, e.g., 10^5 WL), the number of mining tickets allocated to provider i is*

$$\textit{Tickets}_i = \min\left(\left\lfloor \frac{S_i}{S_0} \right\rfloor ,\ C_i \right). \tag{2}$$

For VCT, each ticket is associated with a specific Compute Slot DID; a slot may hold at most one ticket at a time, and stake bonded to a provider is automatically distributed among that provider's slots up to the slot count. Surplus stake beyond capacity, or surplus capacity beyond stake, accrues no additional tickets.

C. Sybil Resistance

Proposition 1 (Hybrid Ticket Sybil Bound)**.** *Let an adversary control aggregate stake budget Σ and aggregate physical capacity Γ, distributed across an arbitrary number of Provider DIDs. The total number of mining tickets the adversary can hold is bounded by*

$$\textit{Tickets}_{adv} \ \leq\ \min\left(\left\lfloor \frac{\Sigma}{S_0} \right\rfloor ,\ \Gamma \right). \tag{3}$$

Proof. Equation (2) is monotone in S_i and C_i separately, and the min operation is subadditive: for any partition of Σ and Γ into provider buckets i with $\sum_i S_i = \Sigma$ and $\sum_i C_i = \Gamma$,

$$\sum_i \min(\lfloor S_i/S_0 \rfloor, C_i) \leq \min(\lfloor \Sigma/S_0 \rfloor, \Gamma) + O(\text{partition rounding}),$$

where the rounding correction is bounded above by the number of partitions and is negligible at scale. The adversary cannot exceed the right-hand side. □

Corollary 1. *Stake-only Sybil and hardware-only Sybil attacks are both ineffective: an adversary with stake Σ but capacity $\Gamma = 0$ obtains zero tickets, and an adversary with capacity Γ but stake $\Sigma = 0$ obtains zero tickets. Both stake and capacity must scale together.*

This dual binding is precisely the property pure stake-only or pure hardware-only allocations lack. It also explains why the WorldLand network can offer stake-bound mining without the centralization risks of pure Proof-of-Stake: the capacity term anchors influence in physical infrastructure.

V. Tail Certificates

A. Construction

Definition 3 (Tail Certificate)**.** *For a Compute Slot j controlled by provider i with VRF secret key sk_j and the canonical seed $seed_h$ at block height h (derived from the previous finalized block header per [1]), the slot computes*

$$(y_j, \pi_j) = \mathrm{VRF}\left(sk_j,\ seed_h \,\|\, DID_j \,\|\, Account_i \,\|\, h\right). \tag{4}$$

Let T_h denote the canonical heads threshold for block h (a deterministic function of the protocol parameter p and the block index, see [1]). A Tail Certificate *for slot j at height h is the tuple*

$$\mathrm{TailCert}_j^h \ = \ \left(DID_j,\ Account_i,\ h,\ seed_h,\ y_j,\ \pi_j,\ T_h\right), \tag{5}$$

which is well-formed when $y_j \geq T_h$ (i.e., the VRF output indicates tails).

B. Verification

A relying party verifies $\mathrm{TailCert}_j^h$ by (1) checking that π_j is a valid VRF proof for the registered public key pk_j on input $\mathrm{seed}_h \,\|\, \mathrm{DID}_j \,\|\, \mathrm{Account}_i \,\|\, h$, (2) confirming via the on-chain registry that DID_j is bonded to $\mathrm{Account}_i$ and active in the current epoch, (3) confirming seed_h matches the canonical seed for height h on the relying party's view of the chain, (4) confirming T_h matches the canonical threshold, and (5) confirming $y_j \geq T_h$. All five checks are constant-time given on-chain state.

C. Unforgeability

Proposition 2 (Tail Certificate Unforgeability)**.** *Under the EUF-CMA security of the underlying VRF [2], no probabilistic polynomial-time adversary can produce a valid* $\mathrm{TailCert}_j^h$ *for a Compute Slot j at block height h when the slot's actual VCT outcome at h is heads ($y_j < T_h$), except with negligible probability.*

Proof sketch. Equation in Definition 3 fixes the input to the VRF given seed_h, DID_j, $\mathrm{Account}_i$, and h. Since seed_h is fixed by the previous finalized block, DID_j and $\mathrm{Account}_i$ are fixed by the on-chain registry, and h is the height of the candidate block, the VRF input is uniquely determined. By the deterministic-output property of the VRF, y_j is the unique output for the registered sk_j, and any successful forgery would require either (a) producing a valid VRF proof for a different $y'_j \neq y_j$ on the same input, contradicting EUF-CMA security, or (b) controlling a different sk'_j corresponding to the same registered pk_j, contradicting the security of the digital signature scheme underlying the registry binding. □

D. Marketplace Integration

The on-chain workload broker enforces a gate at the workload assignment boundary:

A Compute Slot whose VCT outcome at h is heads cannot produce a verifying $\mathrm{TailCert}_j^h$ (Proposition 2). The marketplace will therefore not assign new workloads to that slot at block h, and any active workload lease on that slot will fail the per-block re-certification (Section VI).

Algorithm 1 Workload Assignment Gate

Require: Open job J at block h, candidate slot j

1: Receive TailCert_j^h from provider i controlling slot j
2: **if** TailCert_j^h does not verify **then**
3: Reject the candidacy
4: **else**
5: Allow slot j to bid on or accept job J
6: **end if**
7: Repeat at every block h during the lease lifetime

E. Privacy Properties

A meaningful side-effect of the Tail Certificate construction is that the network's heads set—the set of slots subject to mining duty at block h—is not enumerated in any public location. A passive observer learns only the identities of slots that publish Tail Certificates (and thereby disclaim mining duty); it does not learn the identities of slots selected for mining unless those slots themselves publish Head Certificates (e.g., as part of block submission or participation share). This affords WorldLand a stronger DDoS-resistance posture than committee-based VRF protocols [3] that publish committee membership.

VI. Marketplace Design

A. Mode Selection at Epoch Boundaries

At each epoch boundary, every Compute Slot DID declares one of three modes for the upcoming epoch:

- **Consensus + Spot Service**: the slot is included in the VCT eligible set and may serve preemptible workloads when not selected for mining. This is the default mode for most participants.
- **Service-Only**: the slot is excluded from the VCT eligible set for the epoch and may serve reserved, non-preemptible workloads. The slot earns no mining rewards but commands premium service pricing.
- **Offline / Draining**: the slot is unavailable for both mining and service. Used for maintenance, reconfiguration, or operator-initiated departure.

The mode is recorded on-chain at the start of each epoch. Mid-epoch transitions are forbidden: a slot cannot, upon receiving a heads outcome, attempt to switch to Service-Only mode and avoid mining duty. Per-epoch mode commitment is the temporal counterpart to the per-block Tail Certificate gate; together they prevent both block-level and epoch-level defection.

B. Two Compute Products

The marketplace offers two distinct compute products:

- **Spot Compute**: served by Consensus + Spot Service mode slots. Tenants accept that their workloads may be preempted when their assigned slot is selected for mining. Pricing is lower; suitable for batch inference, rendering, checkpointable training, embarrassingly parallel jobs, and other preemption-tolerant workloads.
- **Reserved Compute**: served exclusively by Service-Only mode slots. No preemption risk; tenants pay premium rates for guaranteed continuity. Suitable for long-running training, real-time inference, and workloads with strict service-level agreements.

The strict separation of the two product classes by underlying slot mode prevents the otherwise-tempting design choice of allowing reserved workloads on consensus slots, which would re-introduce the original mining-defection incentive at the marketplace boundary.

C. Per-Block Lease Semantics

Definition 4 (Tail-Certified Compute Lease)**.** *A lease for a Spot Compute job J assigned to slot j over blocks $[h_0, h_1]$ requires, for each block $h \in [h_0, h_1]$, the submission of a valid* $\mathrm{TailCert}_j^h$ *to the broker prior to expiry of the per-block grace window. Failure to submit triggers immediate suspension of the lease and forfeiture of payment for blocks not certified.*

The per-block re-certification has three consequences: (i) a slot that becomes heads mid-lease is automatically excluded from the lease for the relevant block, freeing the GPU to perform mining without breach; (ii) the broker can verify continuity of service from on-chain evidence alone, without relying on tenant-side reporting; and (iii) payment release is mechanically tied to consensus participation, so a provider cannot be paid for service revenue during blocks in which the slot defected from mining duty.

D. Payment Escrow

For each Spot Compute lease, the tenant escrows the full lease price V at lease initiation. For each block h covered by the lease, the escrow releases $V/(h_1 - h_0 + 1)$ to the provider conditional on (a) submission of a valid $\mathrm{TailCert}_j^h$, (b) absence of a tenant-initiated dispute within a grace window, and (c) submission of a job heartbeat or output-commitment by the provider. Funds for blocks failing any of these conditions remain escrowed and may be returned to the tenant or contributed to the security pool, per protocol parameter.

E. Fee Currency Design

Both the Service-Only opt-in fee F_S (Section VII) and the marketplace settlement currency are denominated in a fiat-equivalent unit (e.g., a regulated stablecoin or a basket-pegged synthetic), not in WL tokens. This design choice is essential to preserve the deterrent effect of F_S when the WL token's fiat value declines: a fee denominated in WL would lose proportionally to any token depreciation, exactly when the dynamic pricing rule needs maximum bite. Mining rewards remain denominated in WL since they are funded by token issuance and represent the protocol's native security budget.

VII. Reward Structure and Dynamic Security Pricing

A. Three-Part Reward Decomposition

Mining rewards are decomposed into three components for each block h with block reward R_h:

$$R_h = R_h^{\text{winner}} + R_h^{\text{participation}} + R_h^{\text{availability}}, \tag{6}$$

$$R_h^{\text{winner}} = \alpha_W R_h, \quad R_h^{\text{participation}} = \alpha_P R_h, \quad R_h^{\text{availability}} = \alpha_A R_h, \tag{7}$$

with $\alpha_W + \alpha_P + \alpha_A = 1$.

Winner reward ($\alpha_W \approx 0.65$): paid to the slot whose ECCPoW solution is included in the canonical block at height h.

Participation share reward ($\alpha_P \approx 0.20$): distributed equally among slots that submitted a valid heads proof and a low-difficulty ECCPoW participation share for height h, regardless of whether they produced the canonical block. This component eliminates the incentive for selected slots to skip mining on the grounds that their probability of winning is low; submitting a participation share is rational at any expected mining-reward level.

Availability reward ($\alpha_A \approx 0.15$): distributed proportionally across all Consensus + Spot Service mode slots that maintained valid attestation and liveness throughout the epoch in which h falls. This component compensates slots for committing to consensus mode rather than the typically more lucrative Service-Only mode.

The exact values of $\alpha_W, \alpha_P, \alpha_A$ are protocol parameters and are governance-adjustable.

B. Service Revenue Redirection

For each completed Spot or Reserved Compute lease with gross revenue V, the protocol distributes

$$V = V^{\text{provider}} + V^{\text{security}} + V^{\text{operations}}, \tag{8}$$

with $V^{\text{provider}} = \beta_V V$ (e.g., $\beta_V \approx 0.88$), $V^{\text{security}} = \beta_S V$ (e.g., $\beta_S \approx 0.07$), and $V^{\text{operations}} = \beta_O V$ (e.g., $\beta_O \approx 0.05$).

The security-pool share V^{security} enters a protocol-controlled treasury that funds the availability reward $R_h^{\text{availability}}$ above the level supported by token issuance alone. As marketplace activity grows, the security pool grows with it, and the network's consensus security budget grows in step. This linkage ensures that the marketplace does not cannibalize consensus security but instead reinforces it: every dollar of service revenue contributes to consensus capacity directly via the availability reward channel.

C. Dynamic Security Pricing

Let N_e denote the average number of Consensus + Spot Service mode slots active in epoch e, and let $N_{\min}$ be a protocol-defined floor (e.g., $N_{\min} = 10^4$). When $N_e < N_{\min}$, the protocol enters *security-pricing mode* and adjusts parameters as follows:

- α_A is increased by a factor $\rho(N_e/N_{\min})$, reducing α_W proportionally.
- The Service-Only mode opt-in fee is increased by the same factor, making consensus mode relatively more attractive.
- A surcharge is applied to Reserved Compute lease prices, with the surcharge revenue contributing entirely to the security pool.

The factor ρ is a smooth, monotone-decreasing function with $\rho(1) = 1$ and $\rho(0) = \rho_{\max}$ for a parameter $\rho_{\max}$ (e.g., $\rho_{\max} = 5$). The intuition is straightforward: when consensus capacity is scarce, the protocol prices consensus participation more highly, restoring incentive balance and drawing capacity back into the eligible set.

Proposition 3 (Dynamic Pricing Self-Stabilization)**.** *Let providers respond to changes in expected per-slot revenue by reallocating capacity toward whichever mode (Consensus + Spot or Service-Only) offers higher expected revenue. Then:*

(i) Demand-side perturbations. Under a perturbation to either Spot or Reserved compute demand—e.g., a multiplicative shock to the inverse-demand intercepts—the dynamic security pricing rule drives

the system to a stable equilibrium with N_e^* closer to $N_{\min}$ than the equilibrium reached without dynamic pricing.

(ii) Supply-side perturbations. Under a perturbation to the WL token's fiat value, the rule provides partial decoupling: it bounds the equilibrium gap below $N_{\min}$ but does not eliminate it, since mining rewards are intrinsically denominated in WL.

A formal treatment of the equilibrium under specific demand assumptions is reserved for follow-up work; we sketch the argument here. For $N_e < N_{\min}$, $\rho > 1$ raises consensus rewards (via α_A), increases the Service-Only opt-in fee, and applies a surcharge to Reserved Compute lease prices. The first two effects on their own are of moderate magnitude—α_A is a small fraction of the total reward, and F_S is a small fraction of π_S—but the third effect is structurally large because the surcharge directly reduces π_S and is denominated in fiat. The combined effect makes consensus mode strictly more attractive, drawing capacity back into the eligible set. Empirical validation of this behavior is presented in Section VII-D.

D. Empirical Validation: Agent-Based Simulation

To validate Proposition 3 and identify the operational range of the dynamic pricing rule, we perform an agent-based simulation of the provider population under three scenarios: a baseline with no shock, a 50% drop in WL token fiat value, and a 2× surge in Reserved compute demand. In each simulation, $N = 2000$ providers with heterogeneous capacities (Pareto-distributed, shape $\alpha = 1.5$, clipped to $[1, 30]$ slots) and heterogeneous opportunity costs ($\theta_i \sim \mathcal{N}(2.0, 1.0^2)$, clipped to $[0.1, 10]$) update their mode choices asynchronously by best response with a switching margin of 1.5 fiat-equivalent units, with adjustment fraction 0.08 per epoch over 250 epochs. Parameters $\alpha_W = 0.65$, $\alpha_P = 0.20$, $\alpha_{A,\text{base}} = 0.15$, $N_{\min} = 3000$, and $\rho_{\max} = 5$ are used throughout, with the surcharge intensity set so that $V_{\text{reserved}}^{\text{net}} = (1 - (\rho - 1) \cdot \sigma)V_{\text{reserved}}$ for $\sigma = 1$.

Figure 1 shows convergence at the baseline. Both with and without dynamic pricing, N_e settles near $N_e^* \approx 3400 > N_{\min}$, indicating that the natural equilibrium under nominal parameters has positive headroom over the security floor; the dynamic pricing rule is dormant in this regime, as designed.

Figure 2 shows the response to a 2× Reserved-demand surge at epoch 50. Without dynamic pricing, capacity migrates aggressively from Consensus toward Service-Only mode, and the post-surge equilibrium settles at $N_e \approx 1271$, far below the security floor and representing a 58% capacity loss. With dynamic pricing active, the post-surge equilibrium settles at $N_e \approx 2274$—a 79% improvement in retained consensus capacity, with the gap from $N_{\min}$ reduced from 58% to 24%.

Under a 50% WL token price drop (not shown in figure), the corresponding equilibria are $N_e \approx 2615$ (without dynamic pricing) and $N_e \approx 2826$ (with dynamic pricing), an 8% improvement. The smaller magnitude reflects the supply-side limitation in Proposition 3(ii): mining rewards are denominated in WL and cannot be fully insulated from token-side shocks via in-protocol mechanisms alone. Mitigating supply-side shocks beyond this point would require cross-economy interventions (treasury operations, emergency issuance) outside the scope of the present mechanism.

Figure 3 confirms that the dynamic pricing effect is robust across provider heterogeneity levels: under demand surge, the relative improvement of dynamic over static pricing ranges from +54% in highly heterogeneous populations to +109% in homogeneous populations, confirming that the mechanism does not rely on fine-tuned demographic assumptions.

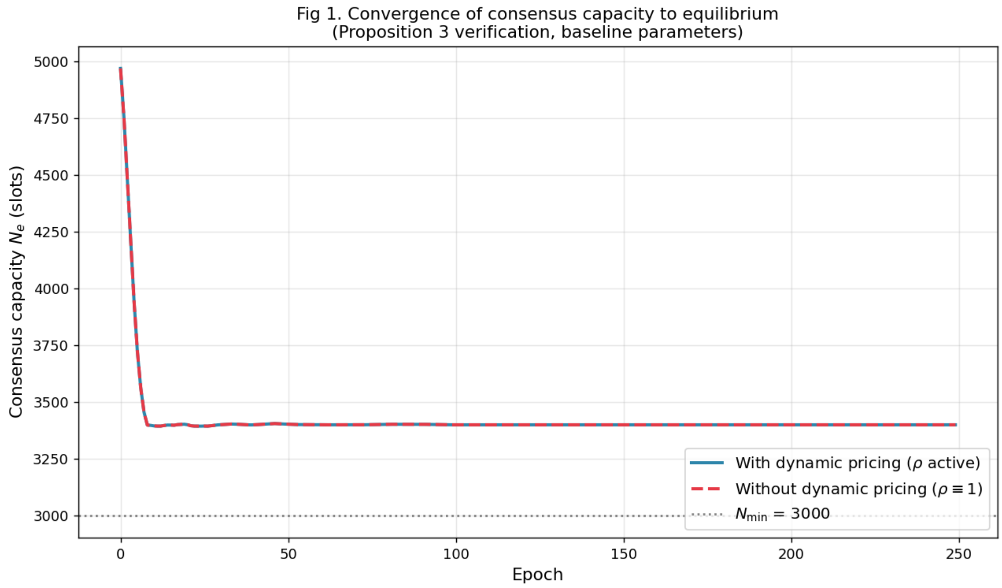

Fig. 1. Baseline convergence of consensus capacity N_e. With and without dynamic pricing, the system settles near $N_e^* \approx 3400$, above $N_{\min} = 3000$, after an initial migration from full-consensus initialization. Dynamic pricing has no effect when $N_e > N_{\min}$, by construction.

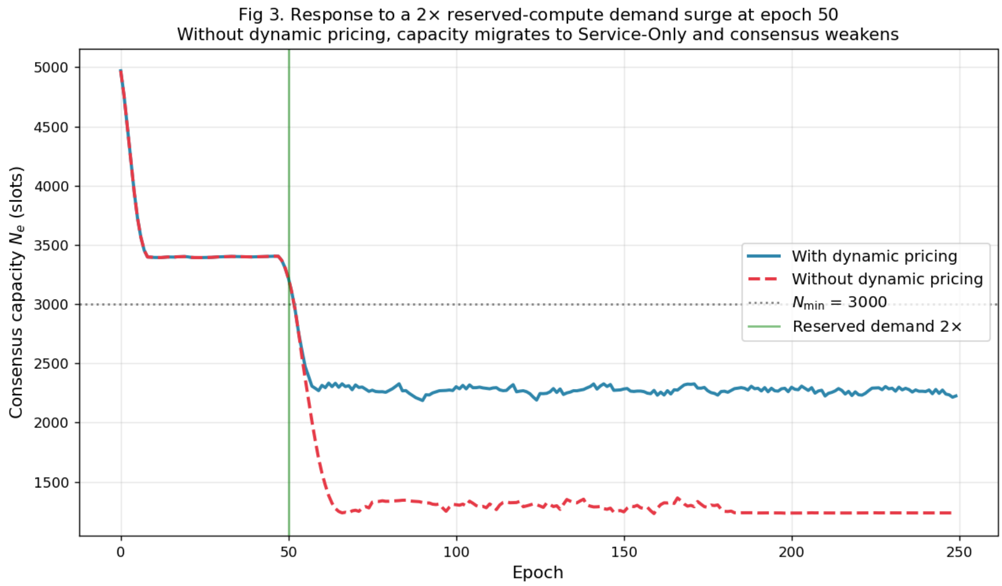

Fig. 2. Response to a 2× Reserved-demand surge at epoch 50. Without dynamic pricing, N_e collapses to 1271 (42% of $N_{\min}$). With dynamic pricing, N_e stabilizes at 2274 (76% of $N_{\min}$)—confirming Proposition 3(i) for demand-side perturbations.

E. Calibration of $\rho_{\max}$

The cap $\rho_{\max}$ on the dynamic pricing factor governs the maximum subsidy the security pool can extend to consensus participants in scarcity. To inform its calibration, we sweep $\rho_{\max} \in \{1.0, 1.5, 2.0, 3.0, 5.0, 10.0\}$ under the same three scenarios and report the resulting post-shock equilibrium in Figure 4.

Two observations follow:

(i) Adequacy of moderate caps. Within the shock magnitudes simulated, $\rho_{\max} = 1.5$ is sufficient to extract the full benefit of dynamic pricing. The plateau in Figure 4(a) for $\rho_{\max} \geq 1.5$ indicates that

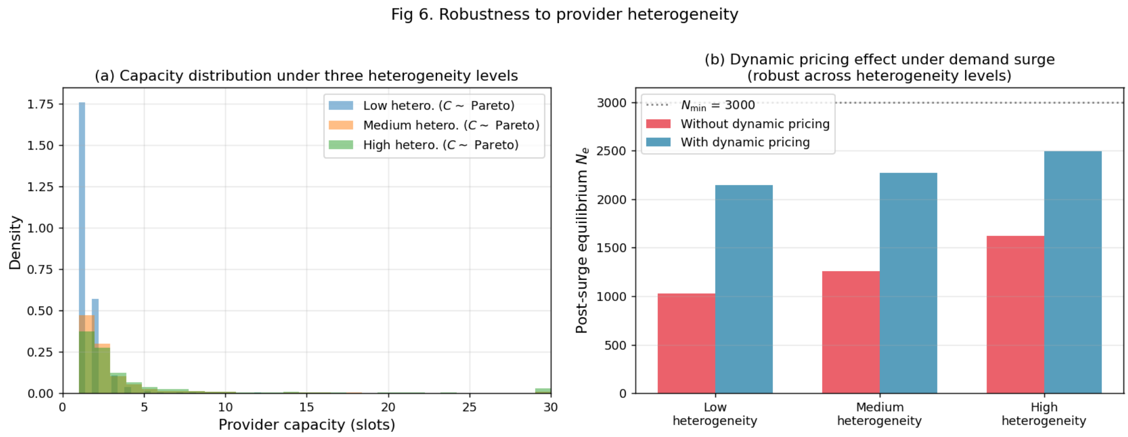

Fig. 3. Robustness to provider heterogeneity. Three Pareto shape parameters and three opportunity-cost spreads define low, medium, and high heterogeneity regimes. The dynamic pricing improvement under demand surge is positive and substantial (+54% to +109%) across all three regimes.

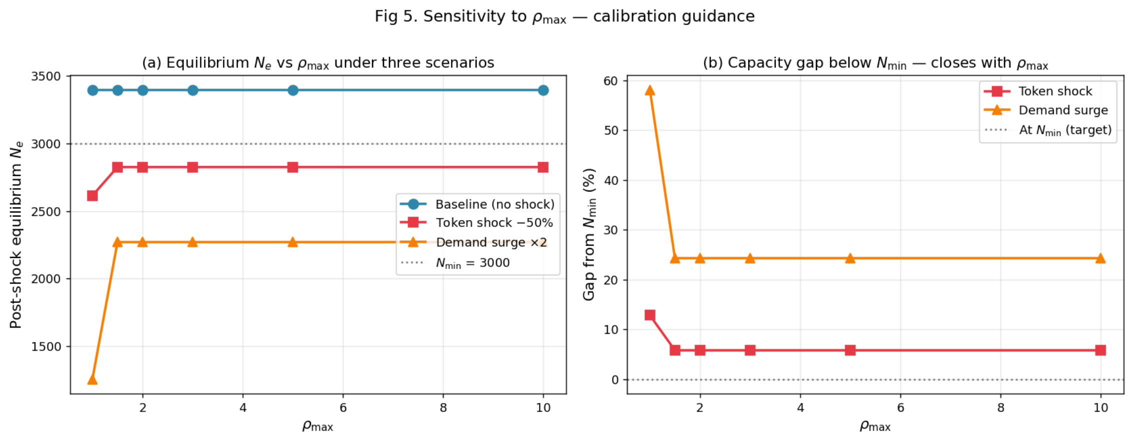

Fig. 4. Sensitivity to $\rho_{\max}$. Equilibrium N_e improves sharply between $\rho_{\max} = 1$ and $\rho_{\max} = 1.5$, then plateaus, indicating that the realized ρ value at equilibrium under these specific shocks ranges from approximately 1.05 to 1.30. For more severe tail shocks, larger $\rho_{\max}$ caps would be required.

the realized ρ at equilibrium remains below 1.5; further headroom does not change the equilibrium because it is not exercised.

(ii) Tail-risk reserve. For shocks substantially larger than those simulated—e.g., an 80% token price drop, a 5× Reserved demand surge, or simultaneous demand and supply shocks—larger $\rho_{\max}$ would be required to maintain N_e near $N_{\min}$. We recommend $\rho_{\max} \geq 5$ as a conservative protocol default, providing reserve capacity for tail conditions without operational cost during normal operation.

The dominant mechanism among the three components of the dynamic pricing rule is the Reserved-compute surcharge. Both α_A scaling and F_S scaling provide moderate effects bounded by their respective base values, but the surcharge directly reduces the right-hand side of the provider's mode-choice inequality $\pi_C \geq \pi_S$ and is denominated in fiat-equivalent units, escaping the WL price exposure that limits the other two components. Calibration of the protocol should therefore prioritize the surcharge intensity parameter.

VIII. Provider Strategy Analysis

A. Rational Provider Game

We sketch the strategic situation faced by a rational provider deciding whether and how to participate in the protocol, taking the marketplace state and reward parameters as exogenous.

The provider's epoch-level decision is the partition of capacity across the three modes (Consensus + Spot, Service-Only, Offline). Once Consensus + Spot capacity is committed, the per-block strategic choice—whether to perform mining duty when selected—has been collapsed to a non-issue by the Tail Certificate mechanism: the only way to earn marketplace revenue during blocks h for which a slot is heads is to also earn no mining reward, which is strictly dominated by performing the mining and receiving the participation share reward. The provider's strategic action space therefore reduces to the epoch-level partitioning decision.

B. Off-Protocol Defection

A provider with control over their own physical infrastructure can, in principle, run private workloads off-protocol on a slot that has been declared in Consensus + Spot mode. If selected for mining at block h, the provider could attempt to continue the off-protocol workload while ignoring mining duty.

This defection has two consequences. First, the provider forfeits the participation share reward for block h, which represents a small but positive loss in expectation. Second, repeated defection eventually triggers a reputation degradation in the on-chain provider registry, reducing future workload assignment from the in-protocol marketplace. The combined cost is bounded but non-trivial; for providers whose primary revenue source is the in-protocol marketplace, off-protocol defection is dominated by mining duty fulfillment in equilibrium.

We do not claim off-protocol defection is impossible—it is fundamentally a physical-control question that no on-chain mechanism can fully address. We do claim that the in-protocol marketplace's dependence on Tail Certificates removes the economic incentive to defect, which is the practically tractable form of the problem.

C. Token Price Sensitivity

If the WL token price falls, mining rewards (denominated in WL) become less attractive in fiat terms relative to service revenue (typically denominated in stablecoin or fiat-equivalent). Without the dynamic security pricing rule, a sustained WL price decline could induce capacity migration from Consensus to Service-Only, reducing N_e below $N_{\min}$ and degrading consensus security.

The dynamic security pricing rule (Section VII) is the protocol's principal defense against this scenario. As N_e declines, α_A rises, the Service-Only opt-in fee rises, and the Reserved Compute surcharge directs marketplace revenue—which is denominated in fiat-stable terms—toward the security pool. This converts marketplace revenue into consensus security at a rate that increases as security becomes scarcer, partially decoupling consensus capacity from WL token price.

IX. Implementation and Roadmap

A. Relation to the Coin-Toss Protocol Roadmap

This protocol is designed to integrate with the implementation roadmap described in [1], which organizes the Rockies Upgrade series into three concurrent tracks. The Tail-Certificate Protocol activates as part of Track C and is targeted for the **Rockies Upgrade v.1.2** milestone (June 2027). Specifically:

- Hierarchical compute identity (Section III) and the hybrid ticket model (Section IV) are deployed in the same block as the Tail Certificate primitive, since they are pre-conditions for slot-level VCT and ticket allocation.
- The marketplace and reward design (Sections VI–VII) are activated in the v.1.2 hardfork.
- The dynamic security pricing rule operates from v.1.2 onward, with the floor $N_{\min}$ and the function ρ initialized to conservative defaults and adjusted by governance vote thereafter.

The earlier Rockies Upgrade milestones (v.1.0 mainnet migration, v.1.1 hardware-anchored identity layer) provide the consensus and identity foundations on which the present protocol depends, and are described in [1].

B. *Engineering Milestones for v.1.2*

A condensed engineering plan for the Tail-Certificate Protocol activation:

1) **Compute slot registry contract** (Q4 2026): on-chain registry for the three-level identity hierarchy, with slot-level capacity weights and tier classification.
2) **Tail Certificate verification library** (Q4 2026): efficient verification primitive for use by the workload broker; protocol-level inclusion in the consensus client.
3) **Workload broker** (Q1 2027): on-chain orderbook and assignment logic with the per-block Tail Certificate gate.
4) **Reward distribution upgrade** (Q1–Q2 2027): split block reward distribution across winner, participation, and availability components; introduction of the security pool treasury.
5) **Service revenue redirection** (Q2 2027): protocol-level fee splitting on settled marketplace transactions.
6) **Dynamic pricing controller** (Q2 2027): on-chain monitoring of N_e, computation of ρ, and adjustment of α_A, opt-in fees, and surcharges.
7) **Hardfork activation** (June 2027): coordinated activation as Rockies Upgrade v.1.2.

X. Related Work

Decentralized GPU marketplaces—Akash [9], Render [10], Clore [11], Golem [12], io.net [13]—provide useful comparison points but do not address the dual-mode mining/service tension because their networks lack a Proof-of-Work consensus layer drawing on the same hardware. Clore's design, in which idle marketplace capacity automatically falls back to mining, is structurally similar to a one-sided version of the dual-mode model but does not impose mandatory mining duty when capacity is rented; its mining is a backstop, not a security primitive.

Mechanism design for blockchain economies has been studied in the context of MEV [7], fee markets [8], and validator incentives in PoS [6], but the specific problem of enforcing mandatory consensus participation in the presence of a coexisting workload marketplace is, to our knowledge, new. Closest in spirit is the literature on rational ignorance and validator-selection games in delegated PoS systems, but those frameworks do not feature physical compute as a distinct economic resource.

Hardware attestation and confidential computing [5], [4] provide the foundation on which the Tier A and Tier B identity classes are built; we draw on this stack but do not propose new attestation primitives. The Compute Slot DID, the hybrid ticket model, the Tail Certificate, and the marketplace-revenue redirection are the protocol's distinguishing contributions.

XI. Conclusion

The Tail-Certificate Protocol resolves the central economic vulnerability of dual-mode blockchain networks: the temptation for providers to defect from mining duty when service revenue is more lucrative. By making valid Tail Certificates a precondition for marketplace revenue, the protocol converts the per-block mining/service decision into a non-decision: a slot selected for mining cannot earn marketplace revenue at that block, and its only path to in-protocol revenue is to perform the mining duty. The hybrid ticket model bounds Sybil attacks across both stake and physical capacity; the hierarchical compute identity scheme operates at GPU granularity; the three-part reward structure ensures positive expected value for participation regardless of block-winning probability; and the dynamic security pricing rule maintains consensus capacity in the face of changing marketplace conditions.

Together with the consensus protocol of [1], the present protocol completes the architectural foundation for the Rockies Upgrade era of WorldLand. Implementation is targeted for the v.1.2 hardfork in mid-2027.

Future work will address: (i) a formal game-theoretic treatment of the equilibrium under heterogeneous provider preferences and stochastic workload demand, building on the sketch of Proposition 3; (ii) empirical calibration of $N_{\min}$, $\alpha_W, \alpha_P, \alpha_A$, $\beta_V, \beta_S, \beta_O$, and ρ from beta-network operational data; (iii) extension of the Tail Certificate construction to non-blocking attestation refresh, allowing sub-block-time job preemption; and (iv) integration of confidential AI workload pricing for Tier A slots, leveraging GPU attestation for verifiable workload execution.

Acknowledgments

The protocol design presented here arose from a sustained design dialog supported in part by AI-assisted critique. The reference implementation will be developed by graduate students of the INFONET Laboratory at GIST as part of the Track C engineering plan summarized in Section IX, with senior review and refinement following the same governance procedures applied to [1]. The author thanks the WorldLand engineering team and the broader DePIN community for the operational and economic discussions that shaped the marketplace design.

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